Victorine Younang, Ph.D.

Researcher & Educator (ML / AI) — Ph.D., Computer Science

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SUMMARY

Computer scientist and educator (Ph.D., Computer Science) specializing in machine learning, generative AI, and secure/trustworthy AI. Sole-instructor teaching experience (30+ students) across the CS / data-science / security curriculum, a peer-reviewed research program (IEEE TDSC/TSC, with NDSS and AAAI submissions under review), and applied industry impact. Builds rigorous, responsible, and reproducible AI curricula and mentors students from first principles to modern practice.

TEACHING & MENTORSHIP

- **Instructor of Record — CSI 3150, Web & Mobile Systems**, Oakland University (Winter 2026): sole instructor for 30+ students; designed original curriculum, lectures, assignments, and assessments; full-stack (React) and mobile UI/UX.
- **Graduate Teaching Assistant — 12 semesters (2020–2026):** Deep Learning (CSI 5900), Database Systems (CSI 5450), Data Structures (CSI 5610), Computer Networks (CSI 6470), and Senior Capstone (CSI 4240); led labs/recitations, graded, and held office hours.
- Mentored 30+ students; comfortable teaching in person, hybrid, and online.
- **Courses I can teach / develop:** Cloud Security, Computer Forensics, Computer Networks, Data Structures & Algorithms, Databases, Deep Learning, Foundations of Cybersecurity, Generative AI & LLMs, Introduction to Computer Science, Introduction to Data Science, Machine Learning, Machine Learning / AI for Security, Network Administration, Programming, Programming (Python/Java), Responsible & Reproducible AI, Secure Software Development, Statistical Computing (Python/R), Web & Mobile Systems.

RESEARCH

- Active, fundable program in machine learning, data science, and secure AI: Bayesian risk modeling, graph neural networks, neuro-symbolic formal verification, and edge ML.
- Methods: probabilistic graphical models, Bayesian inference, GNN/GAT, LLM fine-tuning, VAR time-series — published in IEEE TDSC/TSC with AAAI/NDSS submissions under review.
- Research naturally enriches data-science teaching with current AI practice.

PROFESSIONAL & APPLIED EXPERIENCE

- **Oakland University — Machine Learning Researcher & Instructor of Record** *(Michigan · Sep 2020 – Apr 2026)*: LLM fine-tuning & evaluation and provable-security research (NSF "Generative AI for Secure Systems"); sole instructor for 30+ students; TA across Deep Learning, Databases, and Networks.
- **Health Alliance Plan (HAP) — Applied Scientist, Clinical NLP** *(Detroit, MI · Summer 2025)*: multimodal BioBERT/ClinicalBERT models predicting 30-day hospital readmissions; SHAP interpretability, bias audits, HIPAA/Snowflake.
- **Rec Room — Applied ML Research Intern** *(Seattle, WA · Summer 2023)*: productionized a graph recommender (+33% session time); real-time inference (–10% latency).
- **EDF — ML Scientist, Grid Forecasting** *(Paris · Summer 2021)*: spatial-temporal GNNs + probabilistic (quantile/Pinball) forecasting of grid load and renewables.
- **EDF — Big Data & ML Engineer** *(Paris · 2019–2020)*: GDPR-compliant data pipelines on Hadoop; –30% downstream errors.
- **Orange Telecom — Data Science / Network Security** *(Paris & Brussels · 2017–2019)*: anomaly/fraud detection on telemetry; secure data pipelines (Kudu/Elasticsearch); R-Impala package.

SELECTED PROJECTS

- **RepoSage — agentic, execution-verified code-&-docs RAG** (LangGraph, vLLM, Qdrant, DPO): dual-source RAG that generates code, executes it in a sandbox, and self-corrects against the library's own tests; self-hosted Qwen2.5-Coder (4-bit); DPO from execution-feedback preferences; RAGAS/Pass@1 evaluation.
- **CipherGuard — reference-free backdoor detection** (NDSS 2027): flags test-vector-compliant backdoors via a data-oblivious invariant; 44 verified ciphers, zero false positives; AST + GNN(GAT) with an Isabelle/HOL-verified corpus of 414 labeled implementations.
- **CryptoGraph — graph ML for structural classification** (GAT+MLP, PyTorch Geometric; IEEE TIFS, in prep): hybrid GNN over AST graphs + global features; F1 0.94, beating GCN, Transformer-Conv, RF, and XGBoost.
- **Bayesian Attack Graphs with Complex Probabilities**: uses complex probabilities to quantify attack likelihood/impact and model destructive interference + backtracking cycles; CVSS v3.1-based; validated on a CAV network.

SELECTED PUBLICATIONS

- V. Younang et al. "Security Risk Assessment with Bayesian Attack Graphs and Complex Probabilities." *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025.

- V. Younang et al. "Extending IoT Devices' Network Lifetime." *IEEE Transactions on Services Computing (TSC)*, 2025.
- V. Younang et al. "Comparative Analysis of Users' Concerns and Government Policies on Autonomous Vehicles." *Springer ICIOT / LNCS*, 2022.
- V. Younang et al. "CipherGuard: Reference-Free Detection of Backdoors in Cipher Implementations." *Under review, NDSS 2027.*
- V. Younang et al. "Tiered Isomorphic Alignment: Curriculum Fine-Tuning across Six LLMs for Verified Code Translation." *Under review, AAAI 2027.*
- V. Younang et al. "CryptoGraph: Learning Security Properties of Block Ciphers with Graph ML." *In prep (resubmission), IEEE TIFS.*

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** data science, statistics, generative AI & LLMs, responsible / reproducible AI, PyTorch, Hugging Face (Transformers, PEFT), deep learning, PyTorch Geometric, graph neural networks (GNN/GAT), representation learning, NLP, recommendation & ranking, classification, anomaly detection, class-imbalance & small-data methods
- **LLMs & Generative AI:** LLM fine-tuning (LoRA, QLoRA, PEFT), reinforcement learning (RLHF/DPO), instruction tuning, model distillation, Mixture-of-Experts (MoE), curriculum learning, RAG, OpenAI API, vector search (ChromaDB, Qdrant), prompt engineering, agentic systems (LangGraph), custom LLM evaluation, BERT/transformer fine-tuning
- **Research Methods & Statistics:** experiment design & A/B testing, statistical learning, Bayesian inference, probabilistic graphical models, complex probability, VAR time-series forecasting
- **Data & Programming:** Python, SQL/NoSQL, Pandas, NumPy, scikit-learn, ETL pipelines, Spark/Hadoop (big data), large-scale scraping (Selenium/BeautifulSoup), R, Kudu, Elasticsearch
- **AI Safety & Security:** LLM evaluation & safety/robustness metrics, AI red-teaming, dangerous-capability evaluation, adversarial ML, trustworthy & verified AI, secure code generation, backdoor detection, formal verification (Isabelle/HOL), AST static analysis, Bayesian attack graphs & threat modeling (CVSS), OWASP Top 10 for LLMs

CERTIFICATIONS

- **AWS Certified AI Practitioner (AIF-C01)** — expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — expected Dec 2026; transformer architectures, prompt engineering, fine-tuning, NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — the de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated in RepoSage.

EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (conferred Apr 2026). Dissertation: *From Perceived Trust to Provable Security: A Multi-Tiered Framework for the Design, Optimization, and Verification of Resource-Constrained IoT Ecosystems.* Advisor: Dr. Amartya Sen.
- **M.Sc. Big Data & B.Sc. Computer Engineering** — ESIGELEC, France (2015–2019). ECE-evaluated as a US Bachelor's + Master's; US-equivalent GPA 3.87/4.0.

SERVICE, FELLOWSHIPS & FUNDING

- CRA-WP Grad Cohort Fellowship (2022, 2023); CRA-IDEALS Fellowship (2022, 2023).
- STEM4ALL mentor (France, 2021); peer reviewer, 5 manuscripts (2024).
- Primary contributor, NSF proposal "Using Generative AI to Design and Develop Secure Systems and Software."